

James Fecanin

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Research experience

- Run and develop 1D climate and photochemical simulations, contributing to algorithm development, model efficiency, and numerical stability
- Modified and extended a General Circulation Model (SPARC/MITgcm within the ADAM framework) for hot Jupiter atmospheres by implementing non-Earth-like atmospheric processes, particularly a more accurate model for magnetic effects
- Mathematical modelling and analysis of MHD and plasma physics in the low magnetic Reynolds number regime

Education

PhD in Astronomy, Institute for Astronomy, University of Edinburgh, U.K. (2025 – present)

Due for completion Feb 2029

Supervised by Dr. Sarah Rugheimer

- Research focus: Mathematical and computational modelling of climates of exo-Titans, including coupling of atmospheric simulations, uncovering and resolving numerical instabilities, and improving Titan-specific large scale modelling framework

MSc in Mathematical and Theoretical Physics, University of Oxford, U.K. (2024 – 2025)

Grade Attained: Distinction

Supervised by: Dr. Thaddeus Komacek (University of Oxford),
Dr. Hayley Beltz (University of Maryland)

- Research focus: Improved treatment of magnetism in MITgcm for hot Jupiter atmospheres by extending an iterative drag scheme to internal dipoles tilted relative to the axis of rotation
- Paper about this work is in preparation, submission imminent
- Taught modules include: Kinetic Theory, Perturbation Methods, Numerical Linear Algebra, Mathematical Geoscience, Advanced Fluid Dynamics and MHD, Geophysical Fluid Dynamics, Algorithms in Theoretical Physics

BA in Physics, Lincoln College, University of Oxford, U.K. (2021 – 2024)

Grade Attained: First Class

- Chosen modules include: Fluids, Exploring Solar Systems, Lagrangian Mechanics, Complex Analysis, Advanced Quantum Mechanics, General Relativity

Centennial High School, Frisco TX, USA (2017 – 2021)

- Graduated 4th out of class of about 550 students

Educational awards

- Scholarship for Master's examinations, Lincoln College.....2025
- Scholarship for part A and B examinations, Lincoln College.....2023 & 2024
- Commendation for practical work, Oxford physics teaching laboratories.....2022 & 2023
- Most outstanding physics student prize, Centennial High School.....2020

Work Experience

Teaching Assistant, School of Physics and Astronomy, University of Edinburgh, U.K. (2026)

- Delivered small group teaching for 2nd year Dynamics and Vector Calculus course
- Demonstrated trickier problems and techniques to larger groups
- Provided one-on-one guidance during interactive problem solving sessions

Internship: Particle Physics Detector Design, University of Oxford, U.K. (Summer 2023)

Supervised by: Prof. Todd Huffman, Dr. Samuel Henry (University of Oxford)

- Performed 8 weeks of computational work modelling the radiation environment of the proposed Electron-Ion Collider at Brookhaven National Laboratory.
- Simulated particle collisions using Geant4, and developed macros tracking particle motion and quantifying radiation deposition into detector components.
- Presented work at collaboration meetings and wrote an internal report to help with detector design.

Independent tutoring in physics and mathematics, online (2019 – 2021, Summer 2024)

- Tutored students in A-level - equivalent physics and mathematics, and provided advice to US students on applying for physics and maths degrees in the UK, including revising for the Oxford PAT and MAT exams.

Technical Skills

Programming: Fortran , Python, C++, MATLAB

Simulation experience: GCMs (MITgcm), MCMC algorithms, 1D climate and photochemical atmospheric models

Computing: experience using HPCs and clusters, and with Windows, MacOS, and Linux operating systems.

Other: LaTeX, HTML
